



Installation Instruction

Installation Instruction Number: 4908413

Released Date: 07-nov-2014

Dual Fuel Build Quality Sign-Off Sheet

Dual Fuel Build Quality Sign-Off Sheet

For each dual fuel system installation, use the following instructions to correctly install all components **before** system commissioning.

Process Instructions

1. Before installing any dual fuel system components, the installing distributor technician **must** complete the Base Engine Preparation process step.
2. Use the separate QSK50 Tier 2 Dual Fuel Installation Instructions to correctly attach tall dual fuel kit components to the QSK50 base engine.
3. Use this sheet to make sure QSK50 Tier 2 Dual Fuel Installation Instructions has been followed.
4. The installing distributor technician will fill out all information required on this sheet.
5. The acting distribution build manager shall sign off on this sheet where notified.
6. The acting distribution build manager shall keep a completed copy of each sheet at the distributor for the life of the engine.

Base Engine Preparation		
Build Information		Start Date:
Engine Serial Number (ESN)		
Installing Distributor		
Installing Technician		
Trailer Manufacturer		
Trailer Number		
Customer		
Engine Rating		
	Complete	Additional Comments
Run Engine At High Idle		
Check for and Troubleshoot Active Fault Codes		
Check for and Troubleshoot Active Engine Derates		
>7000 Hours: Operate Engine Under Load		

	Complete	Additional Comments
Check Injector Condition - No Hot/Cold Cylinders		
Make Sure Air Filters are Clean		
Check Controller Area Network (CAN) Bus		
60 Ohms Resistance Across J1939 BackBone	Ohms Value:	
Process Step Sign-Off	DFSEc or Service Manager Sign-Off	
Kit Assembly	Complete	Additional Comments
Assembly Kit Components		
Dual Fuel Component Arrows Point Downstream of Gas Flow		
Dual Fuel Switches Oriented Correctly		
Dual Fuel Switches Set Correctly		
Regulator Outlet Pressure Switch: 889 mm-H ₂ O [35 in-H ₂ O]		
Fuel Shutoff Valve Inlet Pressure Switch: 1397 mm-H ₂ O [55 in-H ₂ O]		
Balance Lines Installed Without Interference (Avoid Rubbing)		
34 kPa [5 psi] Warning Sticker on Filter Cover		
Knock Sensors Oriented Downward (Connector Pointing Down)		
Unused Gas Train Bungs Plugged		
All Flanges Clear of Tape and Debris		
Installation of Wire Harness		
Gas Train Wire Harness Properly Connected		

Kit Assembly	Complete	Additional Comments
Slack in Wiring Harness Connected to ComAp Panel		
Slick in Wiring Harness Connected to Gas Train Components		
On-Engine Harness Secured to Rear of Engine With P-Clips		
Installation of Diesel Oxidation Catalyst (DOC)		
Make sure Temperature Sensor Wire Harnesses do not Touch the DOC body		
Make sure Temperature Sensors are Correctly Connected (Right, Left, Outlet)		
System Leak Checking		
Fuel System Leak Down: 10 Minutes, No Pressure Loss: 34 kPa [5 psi] $\pm$ 3 kPa [$\pm$ 0.5 psi]		
Fuel Shutoff Valve LEDs Illuminate When Valve Powered ON		
Fuel Shutoff Valve 1: No More Than 11 Bubbles in 10 Seconds		
Fuel Shutoff Valve 2: No More Than 11 Bubbles in 10 Seconds		
Build Aesthetics		
Paint (Covered to Distributor/Original Equipment Manufacturer (OEM) specifications, No Chips or Drips		
OEM/Distributor Bracketry		
Welds		
Process Step Sign-Off	DFSEc or Service Manager Sign-Off	

Final Quality Checks	Complete	Additional Comments
Installation of Dual Fuel Engine Control Module (ECM) Calculations		
ECM Image Saved Before Recalibration		
Parent 0	Before _____ After _____	
Child 1	Before _____ After _____	
Child 2	Before _____ After _____	
ECM Image Saved After Recalibration		
Installation of Panel Configuration File		
Correct Calculation is Loaded on Dual Fuel Panel		
Installation of Dataplate Sticker		
Dual Fuel Dataplate Sticker Installed Over existing "Important Engine Information" Section of Dataplate		
CAN Bus		
600ohms Resistance Across J1939 Backbone	Ohms Value:	
System Verification at Engine High Idle		
Verify Camshaft Position Sensor Light Flashing on Knock Module		
Connect to Panel With InterliMonitor (DriveMonitor)		
Check for Hydraulic Power KW - Signal		
Check for ECM Parameter Fuel Rate		
Check for ECM Parameter Coolant Temperature		
Check for ECM Parameter Engine Speed		
Check the Panel Alarms		

Final Quality Checks	Complete	Additional Comments
Sd LowGasPressIn is Active		
Troubleshoot All Other Active Panel Alarms		
Dual Fuel Kit Warranty and QuickServe® Online Part Updated		
Fill Out and Submit Engine Modification Form		
Warranty Administration Manual - Section 13 Engine History - (111-513-009) Reporting Engine Modifications		
Add Dual Fuel Option Onto Existing ESN Through QuickServe® Online		
Enter ESN QSOL > "Add Upfit" > Enter Dual Fuel Option "CN6001"		
Dual Fuel Kit Installation Complete Date	Date:	
Process Step Sign-Off	DFSEc or Service Manager Sign-Off	
Build Deviations and Additional Checks		
Noted Design Addition, Change, or Issue	Resolution	
Process Step Sign-Off	DFSEc or Service Manager Sign-Off	

Document History

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2015-7-6	Update tables

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